STPM Physics · Semester 3

FORMULA LIST

Oscillations, Waves & Modern Physics · Chapters 20–26. 166 formulas with symbols, units and conditions. Numbered chapter by chapter, taken from the PhysiQ notes.

CHAPTER 20 — OSCILLATIONS

20.1 Characteristics of Simple Harmonic Motion

1	**Defining equation of SHM**	$a = -\omega^2 x$	a = acceleration (m s^{-2}); ω = angular frequency (rad s^{-1}); x = displacement (m)
2	**Restoring force**	$F = -kx = -m\omega^2 x$	k = force constant (N m^{-1}); m = mass (kg); x = displacement (m)
3	**Angular frequency**	$\omega = 2\pi f = \frac{2\pi}{T}$	f = frequency (Hz); T = period (s)
4	**Maximum velocity**	$v_{max} = \omega A$	At x = 0; A = amplitude (m)
5	**Maximum acceleration**	$a_{max} = \omega^2 A$	At x = ±A; A = amplitude (m)

20.2 Kinematics of Simple Harmonic Motion

6	**Displacement**	$x = A \sin \omega t$; $x = A \cos \omega t$	A = amplitude (m); sine form starts at equilibrium; cosine form starts at the extreme position
7	**Velocity**	$v = \frac{dx}{dt} = \omega A \cos \omega t$	v = velocity (m s^{-1})
8	**Acceleration**	$a = -\omega^2 A \sin \omega t = -\omega^2 x$	a = acceleration (m s^{-2})
9	**Velocity at displacement x**	$v = \pm\omega\sqrt{A^2 - x^2}$	x = displacement (m); A = amplitude (m)
10	**Maximum values and period**	$v_{max} = \omega A$; $a_{max} = \omega^2 A$; $T = \frac{2\pi}{\omega}$	v_{max} = maximum velocity (m s^{-1}); a_{max} = maximum acceleration (m s^{-2}); T = period (s)

20.3 Energy in Simple Harmonic Motion

11	**Total energy**	$E = \frac{1}{2}m\omega^2A^2 = \frac{1}{2}kA^2$	m = mass (kg); ω = angular frequency (rad s^{-1}); A = amplitude (m); k = $m\omega^2$ (N m^{-1})
12	**Kinetic energy**	$E_k = \frac{1}{2}mv^2 = \frac{1}{2}m\omega^2(A^2 - x^2)$	E_k = kinetic energy (J); v = velocity (m s^{-1}); x = displacement (m)
13	**Potential energy**	$E_p = \frac{1}{2}m\omega^2x^2 = \frac{1}{2}kx^2$	E_p = potential energy (J); x = displacement (m)
14	**Conservation of energy**	$E_k + E_p = E$	E is constant (J)
15	**Fractions of total energy**	$\frac{E_k}{E} = 1 - (\frac{x}{A})^2$; $\frac{E_p}{E} = (\frac{x}{A})^2$	Ratios of kinetic and potential energy to the total energy; x = displacement (m); A = amplitude (m)

20.4 Systems in Simple Harmonic Motion

16	**Mass–spring period**	$T = 2\pi\sqrt{\frac{m}{k}}$	T = period (s); m = mass (kg); k = force constant (N m^{-1})
17	**Mass–spring angular frequency**	$\omega = \sqrt{\frac{k}{m}}$	ω = angular frequency (rad s^{-1}); k = force constant (N m^{-1}); m = mass (kg)
18	**Simple pendulum period**	$T = 2\pi\sqrt{\frac{L}{g}}$	L = length from the point of suspension to the centre of the bob (m); g = acceleration of free fall (m s^{-2})
19	**Springs in parallel**	$k = k_1 + k_2$	k = effective force constant (N m^{-1}); k_1, k_2 = force constants of the springs (N m^{-1})
20	**Springs in series**	$\frac{1}{k} = \frac{1}{k_1} + \frac{1}{k_2}$	k = effective force constant (N m^{-1}); k_1, k_2 = force constants of the springs (N m^{-1})
21	**Proportionalities for period**	$T \propto \sqrt{m}$; $T \propto \sqrt{L}$; $T \propto \frac{1}{\sqrt{g}}$; $T \propto \frac{1}{\sqrt{k}}$	T = period (s); m = mass (kg); L = length (m); g = acceleration of free fall (m s^{-2}); k = force constant (N m^{-1})

20.5 Damped Oscillation

22	**Amplitude decay**	$A = A_0e^{-bt}$	A_0 = initial amplitude (m); b = damping constant (s^{-1}); t = time (s)
23	**Energy decay**	$E \propto A^2$; $E = E_0e^{-2bt}$	E = energy (J); E_0 = initial energy (J); b = damping constant (s^{-1}); t = time (s)

24	**Exponential decay of amplitude**	$\frac{A_2}{A_1} = \frac{A_1}{A_0}$	For equal time intervals the amplitude falls by the same factor (exponential decay); A_0, A_1, A_2 = amplitudes at successive equal intervals (m)
25	**Half-life of amplitude**	$t½ = \frac{\ln 2}{b}$	$t½$ = half-life of amplitude (s); b = damping constant (s^{-1})

20.6 Forced Oscillation and Resonance

26	**Natural frequency, mass–spring**	$f_0 = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$	f_0 = natural frequency (Hz); k = force constant (N m^{-1}); m = mass (kg)
27	**Natural frequency, simple pendulum**	$f_0 = \frac{1}{2\pi}\sqrt{\frac{g}{L}}$	f_0 = natural frequency (Hz); g = acceleration of free fall (m s^{-2}); L = length (m)
28	**Resonance condition**	$f_{driving} = f_0$	$f_{driving}$ = frequency of the driving force (Hz); f_0 = natural frequency (Hz)

CHAPTER 21 — WAVE MOTION

21.1 Progressive Waves

1	**Wave speed**	$v = f\lambda$	v = wave speed (m s^{-1}); f = frequency (Hz); λ = wavelength (m)
2	**Wave travelling in +x direction**	$y = A\sin(\omega t - kx)$	y = displacement (m); A = amplitude (m); $k = \frac{2\pi}{\lambda}$ = wave number (rad m^{-1}); $\omega = 2\pi f$ (rad s^{-1})
3	**Wave speed in terms of ω and k**	$v = \frac{\omega}{k} = \frac{\lambda}{T}; f = \frac{\omega}{2\pi}$	v = wave speed (m s^{-1}); ω = angular frequency (rad s^{-1}); k = wave number (rad m^{-1}); T = period (s); f = frequency (Hz)
4	**Phase difference**	$\varphi = \frac{2\pi}{\lambda}\Delta x; \varphi = \frac{2\pi\Delta t}{T}$	φ = phase difference (rad); Δx = path separation (m); Δt = time difference (s); T = period (s)
5	**Wave travelling in −x direction**	$y = A\sin(\omega t + kx)$	Travels in the −x direction; symbols as above

21.2 Wave Intensity

6	**Intensity**	$I = \frac{P}{A}$	I = intensity (W m^{-2}); P = power (W); A = area normal to the wave (m^2)
7	**Intensity of a point source**	$I = \frac{P}{4\pi r^2}$	Point source; r = distance from the source (m); P = power (W)

8	**Intensity and amplitude**	$I \propto A^2$; $\frac{I_2}{I_1} = (\frac{A_2}{A_1})^2$	A = amplitude; I = intensity (W m^{-2})
9	**Inverse square law**	$I_1 r_1^2 = I_2 r_2^2$; $I_2 = I_1(\frac{r_1}{r_2})^2$	I_1, I_2 = intensities at distances r_1, r_2 from the source (W m^{-2}, m)
10	**Amplitude of a spherical wave**	$A \propto \frac{1}{r}$	Spherical wave; A = amplitude; r = distance from the source (m)

21.3 Principle of Superposition

11	**Constructive interference**	$\Delta x = n\lambda$	Δx = path difference (m); λ = wavelength (m); n = 0, 1, 2, ...; coherent in-phase sources
12	**Destructive interference**	$\Delta x = (n + ½)\lambda$	Δx = path difference (m); λ = wavelength (m); n = 0, 1, 2, ...; coherent in-phase sources
13	**Phase difference**	$\varphi = \frac{2\pi}{\lambda}\Delta x$	φ = phase difference (rad); Δx = path difference (m); λ = wavelength (m)
14	**Resultant amplitude**	$A^2 = A_1^2 + A_2^2 + 2A_1A_2 \cos\varphi$	Two waves of amplitudes A_1 and A_2 (m); φ = phase difference (rad)
15	**Intensity of equal waves**	$I = 4I_0\cos^2\frac{\varphi}{2}$	Equal waves; I_0 = intensity of each wave (W m^{-2}); φ = phase difference (rad)
16	**Special cases of phase difference**	$\varphi = \frac{\pi}{2}$: $A = \sqrt{A_1^2 + A_2^2}$; $\varphi = 0$: $A = A_1 + A_2$; $\varphi = \pi$: $A = \lvert A_1 - A_2 \rvert$	A = resultant amplitude (m); A_1, A_2 = amplitudes of the two waves (m); φ in rad

21.4 Standing Waves

17	**Standing wave on string fixed at both ends**	$L = \frac{n\lambda}{2}$; $f_n = \frac{nv}{2L}$	n = 1, 2, 3, ...; v = wave speed on the string (m s^{-1}); L = length (m); λ = wavelength (m); f_n = frequency of the nth mode (Hz)
18	**Fundamental and harmonics**	$f_1 = \frac{v}{2L}$; $f_n = nf_1$	f_1 = fundamental frequency (Hz); f_n = frequency of the nth harmonic (Hz); v = wave speed (m s^{-1}); L = length (m)
19	**Standing wave equation**	$y = 2A\sin(kx)\cos(\omega t)$	$k = \frac{2\pi}{\lambda}$ (rad m^{-1}); A = amplitude of each travelling wave (m); ω = angular frequency (rad s^{-1}); nodes at $kx = n\pi$
20	**Node spacing**	Distance between adjacent nodes = $\frac{\lambda}{2} = \frac{\pi}{k}$	λ = wavelength (m); $k = \frac{2\pi}{\lambda}$ = wave number (rad m^{-1})
21	**Wave speed on a string**	$v = \sqrt{\frac{T}{\mu}}$	Stretched string; T = tension (N); μ = mass per unit length (kg m^{-1})

21.5 Electromagnetic Waves

22	**Wave equation for EM waves**	$c = f\lambda$	c = speed of light in vacuum (3.00×10^{8} m s^{-1}); f = frequency (Hz); λ = wavelength (m)
23	**Field amplitudes**	$E_0 = cB_0$	E_0 = electric field amplitude (V m^{-1}); B_0 = magnetic field amplitude (T); c = speed of light (m s^{-1})
24	**Speed of light**	$c = \frac{1}{\sqrt{\mu_0\varepsilon_0}}$	$\mu_0 = 4\pi\times10^{-7}$ H m^{-1}; $\varepsilon_0 = 8.85\times10^{-12}$ F m^{-1}
25	**Photon energy**	$E = hf = \frac{hc}{\lambda}$	E = photon energy (J); h = 6.63×10^{-34} J s; f = frequency (Hz); λ = wavelength (m)
26	**Time of travel**	$t = \frac{d}{c}$	t = time (s); d = distance (m); c = speed of light (m s^{-1})

CHAPTER 22 — SOUND WAVES

22.1 Propagation of Sound Waves

1	**Wave speed**	$v = f\lambda$	v = wave speed (m s^{-1}); f = frequency (Hz); λ = wavelength (m)
2	**Sound wave equation**	$y = A\sin(\omega t - kx)$	$\omega = 2\pi f$ (rad s^{-1}); $k = \frac{2\pi}{\lambda}$ (rad m^{-1}); wave speed $v = \frac{\omega}{k}$ (m s^{-1}); A = amplitude
3	**Speed of sound and temperature**	$\frac{v_2}{v_1} = \sqrt{\frac{T_2}{T_1}}$	T = absolute temperature (K); v = speed of sound (m s^{-1}); $v \approx 331$ m s^{-1} at 0 °C
4	**Intensity of a point source**	$I = \frac{P}{A} = \frac{P}{4\pi r^2}$; $\frac{I_1}{I_2} = \frac{r_2^2}{r_1^2}$	Point source radiating uniformly; I = intensity (W m^{-2}); P = power (W); A = area (m^2); r = distance from the source (m)
5	**Echo / sonar distance**	$d = \frac{vt}{2}$	d = distance to the reflector (m); v = speed of sound (m s^{-1}); t = total time for the pulse to travel to the reflector and back (s)

22.2 Sources of Sound

6	**Open pipe harmonics**	$f_n = \frac{nv}{2L}$; $\lambda_n = \frac{2L}{n}$	f_n = frequency of nth harmonic (Hz); n = 1, 2, 3, …; v = speed of sound (m s^{-1}); L = length of pipe (m); λ_n = wavelength (m); pipe open at both ends

7	**Closed pipe harmonics**	$f = \frac{v}{4L}, \frac{3v}{4L}, \frac{5v}{4L}, \ldots; \lambda_1 = 4L$	f = frequency (Hz); v = speed of sound (m s^{-1}); L = length of pipe (m); λ_1 = fundamental wavelength (m); pipe closed at one end
8	**Stretched string harmonics**	$f_n = \frac{n}{2L}\sqrt{\frac{T}{\mu}}$	f_n = frequency (Hz); n = 1, 2, 3, ...; L = length of string (m); T = tension (N); μ = mass per unit length (kg m^{-1})
9	**Closed pipe fundamental**	$v = 4fL; L + e = \frac{\lambda}{4}$	v = speed of sound (m s^{-1}); f = frequency (Hz); L = length of air column at first resonance (m); e = end correction (m); λ = wavelength (m)

22.3 Intensity Level of Sound

10	**Intensity level**	$\beta = 10 \log_{10} \frac{I}{I_0}$	β = intensity level (dB); I = intensity (W m^{-2}); $I_0 = 1.0\times10^{-12}$ W m^{-2}
11	**Intensity from intensity level**	$I = I_0 \times 10^{\beta/10}$	I = intensity (W m^{-2}); $I_0 = 1.0\times10^{-12}$ W m^{-2}; β = intensity level (dB)
12	**Difference in intensity level**	$\beta_2 - \beta_1 = 10 \log_{10} \frac{I_2}{I_1}$	β_1, β_2 = intensity levels (dB); I_1, I_2 = intensities (W m^{-2})
13	**N identical incoherent sources**	$\beta = \beta_1 + 10 \log_{10} N$	β = intensity level of N sources together (dB); β_1 = intensity level of one source (dB); N = number of identical incoherent sources
14	**Level difference with distance**	$\beta_1 - \beta_2 = 20 \log_{10} \frac{r_2}{r_1}$	β_1, β_2 = intensity levels (dB) at distances r_1, r_2 (m) from the source; for a point source in free space
15	**Intensity from a point source**	$I = \frac{P}{4\pi r^2}$	I = intensity (W m^{-2}); P = power of the source (W); r = distance from the source (m)

22.4 Beat

16	**Beat frequency**	$f_{beat} = \lvert f_1 - f_2 \rvert$	f_{beat} = beat frequency (Hz); f_1, f_2 = frequencies of the two sources (Hz)
17	**Time between loud beats**	$T_{beat} = \frac{1}{f_{beat}}$	T_{beat} = time between successive loud beats (s); f_{beat} = beat frequency (Hz)
18	**Resultant frequency**	$f_{av} = \frac{f_1 + f_2}{2}$	f_{av} = resultant frequency (Hz); f_1, f_2 = frequencies of the two sources (Hz)
19	**Resultant displacement**	$y = 2A \cos(\pi(f_1 - f_2)t) \sin(\pi(f_1 + f_2)t)$	y = resultant displacement; A = amplitude of each wave; f_1, f_2 = frequencies (Hz); t = time (s); for equal amplitudes A

22.5 Doppler Effect

20	**Moving source, stationary observer**	$f' = \frac{fv}{v \mp v_s}$	f' = observed frequency (Hz); f = source frequency (Hz); v = speed of sound (m s^{-1}); v_s = speed of source (m s^{-1}); upper sign (−) when the source approaches, lower sign (+) when it recedes
21	**Moving observer, stationary source**	$f' = \frac{f(v \pm v_o)}{v}$	f' = observed frequency (Hz); f = source frequency (Hz); v = speed of sound (m s^{-1}); v_o = speed of observer (m s^{-1}); plus when the observer approaches, minus when moving away
22	**General Doppler formula**	$f' = \frac{f(v \pm v_o)}{v \mp v_s}$	f' = observed frequency (Hz); f = source frequency (Hz); v = speed of sound (m s^{-1}); v_o = speed of observer (m s^{-1}); v_s = speed of source (m s^{-1})
23	**Small-speed approximation**	$\frac{\Delta f}{f} \approx \frac{v_{rel}}{v}$	Δf = change in frequency (Hz); f = frequency (Hz); v_{rel} = relative speed of source and observer (m s^{-1}); v = speed of sound (m s^{-1}); for small speeds

CHAPTER 23 — GEOMETRICAL OPTICS

23.1 Spherical Mirrors

1	**Mirror equation**	$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$	u = object distance; v = image distance; f = focal length (all in m or cm, same unit)
2	**Focal length and radius**	$f = \frac{R}{2}$	f = focal length; R = radius of curvature; $f > 0$ for concave, $f < 0$ for convex
3	**Linear magnification**	$m = -\frac{v}{u} = \frac{h_i}{h_o}$	m = magnification (no unit); h_i = image height; h_o = object height; $\lvert m \rvert > 1$ magnified, $\lvert m \rvert < 1$ diminished
4	**Image same size as object**	$u = v = 2f = R;\ m = -1$	u = object distance; v = image distance; f = focal length; R = radius of curvature; real, inverted image
5	**Virtual image**	$v < 0;\ m = -\frac{v}{u} > 0$	v = image distance; u = object distance; m is positive (upright image)

23.2 Refraction at Spherical Surfaces

6	**Refraction at a spherical surface**	$\frac{n_1}{u} + \frac{n_2}{v} = \frac{n_2 - n_1}{R}$	n_1 = refractive index of the medium containing the object; n_2 = refractive index of the other medium; u = object distance; v = image distance; R = radius of curvature (m)
7	**Apparent depth**	Apparent depth = real depth × $\frac{\text{n of observer's medium}}{\text{n of object's medium}}$; water viewed from air: apparent depth = real depth/n	Depths in m; n = refractive index of water for the water-from-air case
8	**Second focal length**	$f_2 = \frac{n_2 R}{n_2 - n_1}$	f_2 = second focal length (m), obtained from $u \to \infty$; n_1, n_2 = refractive indices; R = radius of curvature (m)
9	**First focal length**	$f_1 = \frac{n_1 R}{n_2 - n_1}$	f_1 = first focal length (m), obtained from $v \to \infty$; n_1, n_2 = refractive indices; R = radius of curvature (m)
10	**Lateral magnification**	$m = -\frac{n_1 v}{n_2 u}$	m = lateral magnification (no unit); n_1, n_2 = refractive indices; u = object distance; v = image distance

23.3 Thin Lenses

11	**Thin lens equation**	$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$	u = object distance; v = image distance; f = focal length (all distances in the same unit)
12	**Lensmaker's formula**	$\frac{1}{f} = (n - 1)(\frac{1}{R_1} + \frac{1}{R_2})$	n = refractive index of lens material; R_1, R_2 = radii of the two surfaces, each positive when its centre of curvature lies on the side from which the light emerges (biconvex: both positive); if the form $\frac{1}{R_1} - \frac{1}{R_2}$ is used, R_2 is negative for a biconvex lens
13	**Linear magnification**	$m = -\frac{v}{u}$	m = magnification (no unit); u = object distance; v = image distance
14	**Power of a lens**	$P = \frac{1}{f}$	P = power (dioptre, D); f = focal length (m)
15	**Lenses in contact**	$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$; $P = P_1 + P_2$	f = combined focal length; f_1, f_2 = focal lengths of the lenses; P = combined power (D); P_1, P_2 = powers of the lenses (D)
16	**Real image same size as object**	$u = v = 2f$	u = object distance; v = image distance; f = focal length

17	**Minimum object-image distance**	$u + v = 4f$	Minimum distance between object and a real image is 4f; f = focal length

CHAPTER 24 — WAVE OPTICS

24.1 Huygens's Principle

1	**Wave speed**	$v = f\lambda$	v = wave speed (m s^{-1}); f = frequency (Hz); λ = wavelength (m)
2	**Refractive index**	$n = \frac{c}{v} = \frac{\lambda_0}{\lambda}$	n = refractive index of the medium (no unit); c = speed in vacuum (m s^{-1}); v = speed in the medium (m s^{-1}); λ_0 = wavelength in vacuum (air) (m); λ = wavelength in the medium (m)
3	**Snell's law**	$n_1 \sin\theta_1 = n_2 \sin\theta_2$	n_1, n_2 = refractive indices of the two media; θ_1, θ_2 = angles measured from the normal
4	**Wavelet radius**	$r = vt$	Distance moved by a wavefront in time t, which is the radius of each secondary wavelet; v = wave speed (m s^{-1}); t = time (s)

24.2 Interference

5	**Constructive interference**	$\Delta = n\lambda$	Δ = path difference (m); λ = wavelength (m); n = 0, 1, 2, …
6	**Destructive interference**	$\Delta = (n + ½)\lambda$	Δ = path difference (m); λ = wavelength (m); n = 0, 1, 2, …
7	**Phase difference**	$\varphi = \frac{2\pi}{\lambda} \times \Delta$	φ = phase difference (rad); Δ = path difference (m); λ = wavelength (m)
8	**Intensity and amplitude**	$I \propto A^2$	I = intensity (W m^{-2}); A = amplitude
9	**Maximum and minimum intensity**	$I_{max} = (\sqrt{I_1} + \sqrt{I_2})^2$; $I_{min} = (\sqrt{I_1} - \sqrt{I_2})^2$	Two coherent waves of intensities I_1 and I_2 (W m^{-2})
10	**Maximum and minimum amplitude**	$A_{max} = A_1 + A_2$; $A_{min} = \lvert A_1 - A_2 \rvert$	Amplitudes add; A_1, A_2 = amplitudes of the two waves

24.3 Two-slit Interference Pattern

11	**Fringe spacing**	$x = \frac{\lambda D}{a}$	x = fringe spacing (m); λ = wavelength (m); D = slit-to-screen distance (m); a = slit separation (m); valid for $D \gg a$
12	**Path difference at a point**	$\Delta = \frac{ay}{D}$	Δ = path difference (m); y = distance from the central fringe (m)

13	**Bright fringe position**	$y_n = \frac{n\lambda D}{a}$	y_n = position of the nth bright fringe (m); n = 0, 1, 2, ...
14	**Dark fringe position**	$y_n = \frac{(n + ½)\lambda D}{a}$	y_n = position of the nth dark fringe (m)
15	**Angular position of bright fringe**	$a \sin \theta = n\lambda$	θ = angular position of the nth bright fringe; a = slit separation (m); λ = wavelength (m)
16	**Wavelength and spacing in a medium**	$\lambda' = \frac{\lambda}{n}$; $x' = \frac{x}{n}$	In a medium of refractive index n; λ′ = wavelength in the medium (m); x′ = fringe spacing in the medium (m)

24.4 Interference in A Thin Film

17	**Thin film, one phase change: constructive**	$2nt = (m + ½)\lambda$	Film of index n in air, one phase change at the top surface; t = film thickness (m); λ = wavelength in air (m); m = 0, 1, 2, ...
18	**Thin film, one phase change: destructive**	$2nt = m\lambda$	Film of index n in air, one phase change at the top surface; m = 0, 1, 2, ...
19	**Thin film, no relative phase change: constructive**	$2nt = m\lambda$	Both reflections with a π change, or neither (e.g. coating on glass with $n_{air} < n_{film} < n_{glass}$)
20	**Thin film, no relative phase change: destructive**	$2nt = (m + ½)\lambda$	Both reflections with a π change, or neither (e.g. coating on glass with $n_{air} < n_{film} < n_{glass}$)
21	**Minimum coating thickness**	$t = \frac{\lambda}{4n}$	Minimum coating thickness for destructive reflection of wavelength λ (m); n = refractive index of the coating
22	**Air wedge dark fringes**	$2t = m\lambda$	Air wedge at normal incidence; dark fringes; t = air gap thickness (m); m = 0, 1, 2, ...
23	**Air wedge fringe spacing**	$x = \frac{\lambda}{2\theta}$	Air wedge at normal incidence; x = fringe spacing (m); θ = wedge angle (rad)

24.5 Diffraction by A Single Slit

24	**Single-slit minima**	$a \sin \theta = m\lambda$	m = 1, 2, 3, ... for the minima; a = slit width (m); θ = angle from the central axis (rad or °); λ = wavelength (m)
25	**Angular width of central maximum**	$2\theta_1$; $\sin \theta_1 = \frac{\lambda}{a}$	$2\theta_1$ = angular width of the central maximum (rad); θ_1 = angle of the first minimum; a = slit width (m); λ = wavelength (m)

26	**Width of central maximum on screen**	$w = \frac{2\lambda D}{a}$	small angles; w = width of the central maximum (m); D = distance from slit to screen (m); λ = wavelength (m); a = slit width (m)
27	**Small-angle approximation**	$\theta \approx \frac{\lambda}{a}$	for small angles; θ = angle of the first minimum (rad); λ = wavelength (m); a = slit width (m)

24.6 Diffraction Gratings

28	**Grating equation**	$d \sin\theta = m\lambda$	m = 0, 1, 2, ... (order); d = grating spacing (m); θ = angle of the mth order from the normal (rad or °); λ = wavelength (m)
29	**Grating spacing**	$d = \frac{1}{N}$	N = number of lines per metre (m^{-1}); d = grating spacing (m); for example 500 lines mm^{-1} = 5.00 × 10^5 lines m^{-1}, d = 2.00 × 10^{-6} m
30	**Highest order**	$m_{max} < \frac{d}{\lambda}$	m_{max} = largest integer less than $\frac{d}{\lambda}$ (because sin θ < 1); d = grating spacing (m); λ = wavelength (m)
31	**Total number of maxima**	$2m_{max} + 1$	total number of maxima observed; m_{max} = highest order
32	**Grating angle from screen**	$\tan\theta = \frac{y}{D}$	y = distance on the screen from the central maximum (m); D = distance from grating to screen (m); then find sin θ from θ

24.7 Polarisation

33	**Malus's law**	$I = I_0 \cos^2\theta$	I_0 = intensity of plane-polarised light incident on the analyser (W m^{-2}); I = transmitted intensity (W m^{-2}); θ = angle between the plane of polarisation and the transmission axis
34	**Unpolarised light through a polariser**	$I_1 = \frac{I_{unpolarised}}{2}$	I_1 = intensity after the first polariser (W m^{-2}); $I_{unpolarised}$ = incident intensity (W m^{-2}); then apply Malus's law for each further polariser
35	**Brewster's law**	$\tan\theta_B = \frac{n_2}{n_1}$; $\tan\theta_B = n$	θ_B = Brewster (polarising) angle; n_1, n_2 = refractive indices of the first and second media; $\tan\theta_B = n$ for light incident from air
36	**Reflected and refracted rays at Brewster angle**	$\theta_B + \theta_r = 90°$	θ_B = Brewster angle; θ_r = angle of refraction

24.8 Optical Waveguides

37	**Critical angle**	$\sin\theta_c = \frac{n_2}{n_1}$	θ_c = critical angle; n_1 = refractive index of the core; n_2 = refractive index of the cladding
38	**Numerical aperture**	$NA = \sin\theta_a = \sqrt{n_1^2 - n_2^2}$	θ_a = maximum acceptance angle (from air); NA = numerical aperture (no unit); n_1 = refractive index of the core; n_2 = refractive index of the cladding
39	**Attenuation**	$\text{Attenuation} = \frac{10}{L}\log_{10}\frac{P_{in}}{P_{out}}$	in dB per unit length; L = length of the fibre; P_{in} = input power (W); P_{out} = output power (W)
40	**Pulse transit time along axis**	$t = \frac{L}{v} = \frac{nL}{c}$	t = time for a pulse along the axis (s); L = length of the fibre (m); $v = \frac{c}{n}$ = speed of light in the core (m s^{-1}); n = refractive index of the core; c = speed of light in vacuum (m s^{-1})

CHAPTER 25 — QUANTUM PHYSICS

25.1 Photon

1	**Photon energy**	$E = hf = \frac{hc}{\lambda}$	E = photon energy (J); f = frequency (Hz); λ = wavelength (m); h = 6.63 × 10^{-34} J s; c = 3.00 × 10^8 m s^{-1}
2	**Photoelectric equation**	$hf = \varphi + \frac{1}{2}mv^2_max$; $\varphi = hf_0$	f = frequency of incident radiation (Hz); φ = work function (J); f_0 = threshold frequency (Hz); m = electron mass (kg); v_{max} = maximum speed of photoelectrons (m s^{-1})
3	**Stopping potential**	$eV_s = \frac{1}{2}mv^2_max = hf - \varphi$	V_s = stopping potential (V); e = electron charge (C); m = electron mass (kg); v_{max} = maximum speed of photoelectrons (m s^{-1}); φ = work function (J)
4	**Photon momentum**	$p = \frac{h}{\lambda}$	p = photon momentum (kg m s^{-1}); h = Planck constant (J s); λ = wavelength (m)
5	**Electronvolt**	1 eV = 1.60 × 10^{-19} J; $E\,(eV) \approx \frac{1240}{\lambda}$ (nm)	E (eV) = photon energy in electronvolts; λ (nm) = wavelength in nanometres

25.2 Wave-Particle Duality

6	**de Broglie wavelength**	$\lambda = \frac{h}{p} = \frac{h}{mv}$	λ = de Broglie wavelength (m); h = Planck constant (J s); p = momentum (kg m s^{-1}); m = mass (kg); v = speed (m s^{-1})
7	**Wavelength from kinetic energy**	$p = \sqrt{2mK}$; $\lambda = \frac{h}{\sqrt{2mK}}$	K = kinetic energy (J); m = mass (kg); p = momentum (kg m s^{-1}); λ = wavelength (m)
8	**Electron accelerated through a potential difference**	$eV = \frac{1}{2}mv^2$; $\lambda = \frac{h}{\sqrt{2meV}}$	V = accelerating potential difference (V); e = electron charge (C); m = electron mass (kg); v = speed (m s^{-1}); λ = wavelength (m)

25.3 Atomic Structure

9	**Hydrogen energy levels**	$E_n = -\frac{13.6}{n^2}$ eV	n = 1, 2, 3, … (n = 1 is the ground state); E_n = energy of level n (eV); 1 eV = 1.60×10^{-19} J
10	**Photon emitted or absorbed in a transition**	$hf = \frac{hc}{\lambda} = E_i - E_f$	E_i = energy of the initial (higher) level (J); E_f = energy of the final level (J); f = photon frequency (Hz); λ = wavelength (m)
11	**Rydberg formula**	$\frac{1}{\lambda} = R(\frac{1}{n_f^2} - \frac{1}{n_i^2})$	R = 1.10×10^7 m^{-1} (Rydberg constant); λ = wavelength (m); n_i = initial level; n_f = final level
12	**Bohr radius of orbit**	$r_n = n^2 a_0$	r_n = radius of the nth orbit (m); a_0 = 5.29×10^{-11} m (Bohr radius); n = 1, 2, 3, …

25.4 X-rays

13	**Kinetic energy of electron at target**	$eV = \frac{1}{2}mv^2$	electron reaching the target from rest; V = accelerating potential difference (V); e = electron charge (C); m = electron mass (kg); v = speed at the target (m s^{-1})
14	**Minimum wavelength**	$eV = \frac{hc}{\lambda_{min}}$; $\lambda_{min} = \frac{hc}{eV}$	λ_{min} = minimum wavelength (m); h = 6.63×10^{-34} J s; c = 3.00×10^8 m s^{-1}; e = 1.60×10^{-19} C; V = accelerating potential difference (V)
15	**Bragg's law**	$2d \sin\theta = n\lambda$	d = spacing of the crystal planes (m); θ = glancing angle (measured from the planes, not the normal); n = 1, 2, 3, … (order); λ = wavelength (m)
16	**Attenuation of X-rays**	$I = I_0 e^{-\mu x}$	I_0 = incident intensity (W m^{-2}); I = transmitted intensity (W m^{-2}); μ = linear attenuation coefficient (m^{-1}); x = thickness of the absorber (m)

25.5 Nanoscience

17	**Surface area to volume ratio of a sphere**	$\frac{3}{r}$	r = radius of the sphere (m); ratio in m^{-1}; the ratio increases as the size decreases
18	**Surface area to volume ratio of a cube**	$\frac{6}{L}$	L = side of the cube (m); ratio in m^{-1}; the ratio increases as the size decreases

CHAPTER 26 — NUCLEAR PHYSICS

26.1 Nucleus

1	**Mass defect**	$\Delta m = Zm_p + (A - Z)m_n - m_{nucleus}$	Δm = mass defect (u); Z = proton number; A = nucleon number; m_p, m_n = masses of proton and neutron (u); $m_{nucleus}$ = mass of nucleus (u); use atomic masses with m_H when electrons are included
2	**Binding energy**	$B = \Delta mc^2$; binding energy per nucleon = $\frac{B}{A}$	B = binding energy; Δm = mass defect (u); c = speed of light; 1 u = 931.5 MeV; A = nucleon number
3	**Nuclear radius**	$R = R_0A^{1/3}$	R = nuclear radius (m); $R_0 \approx 1.2\times10^{-15}$ m (1.2 fm); A = nucleon number
4	**Nuclear density**	$\rho = Am_u/(\frac{4}{3}\pi R^3) = \frac{3m_u}{4\pi R_0^3}$	ρ = density (kg m^{-3}); m_u = atomic mass unit (kg); R = nuclear radius (m); $R_0 = 1.2\times10^{-15}$ m; independent of A

26.2 Radioactivity

5	**Decay law**	$N = N_0e^{-\lambda t}$; $A = A_0e^{-\lambda t}$	N = number of undecayed nuclei; A = activity (Bq); N_0, A_0 = values at t = 0; λ = decay constant (s^{-1}); t = time (s)
6	**Half-life**	$t_{½} = \frac{\ln 2}{\lambda} = \frac{0.693}{\lambda}$	$t_{½}$ = half-life (s); λ = decay constant (s^{-1})
7	**Half-life form of decay law**	$N = N_0(½)^{t/t_{½}}$; fraction remaining = $(½)^{t/t_{½}}$	N = number of undecayed nuclei; N_0 = initial number; t = time (s); $t_{½}$ = half-life (s)
8	**Activity**	$A = \lambda N$; $A \propto N$	A = activity (Bq); λ = decay constant (s^{-1}); N = number of undecayed nuclei
9	**Linearised decay law**	$\ln A = \ln A_0 - \lambda t$	A = activity (Bq); A_0 = initial activity (Bq); λ = decay constant (s^{-1}); t = time (s); graph of ln A against t is a straight line with gradient −λ

26.3 Nuclear Reaction

10	**Q-value of a reaction**	$Q = \Delta mc^2 = (\Sigma m_{reactants} - \Sigma m_{products}) \times 931.5$ MeV	Q = energy released (MeV); Δm = mass difference (u); masses in u; c = speed of light
11	**Mass-energy equivalence**	$E = \Delta mc^2$	E = energy (J); Δm = mass difference (kg); c = 3.00×10^{8} m s^{-1}; 1 u = 1.66×10^{-27} kg
12	**Nuclear equation**	ΣA and ΣZ equal on both sides; e.g. $^{2}H + {}^{3}H \rightarrow {}^{4}He + n$	A = nucleon number; Z = proton number; both conserved in a nuclear reaction
13	**Power from fission**	P = (fission rate) × (energy per fission)	P = power (W); fission rate in s^{-1}; energy per fission in J
14	**Mass of ^{235}U used**	$\text{Mass} = \frac{\text{number of fissions} \times 235\ \text{g}}{N_A}$	Mass in g; N_A = Avogadro constant (mol^{-1})